# Reflect with us - Week 4

This assignment will not be graded and is only for practice.
Find the dimension of the vector space

$$V = \{(x, y, z) \mid x, y, z \in R, x + y + z = 0, z = 0\}$$
$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2); \quad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V$$
$$\text{Scalar multiplication: } c(x, y, z) = (cx, cy, cz); \ (x, y, z) \in V, \ c \in R$$

V = {(x, y, z) | x, y, z ∈ R, x + y + z = 0, z = 0}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2); (x1, y1, z1), (x2, y2, z2) ∈ V Scalar multiplication: c(x, y, z) = (cx, cy, cz); (x, y, z) ∈ V , c ∈ R
Solution:Solution:

Observe that, in question, you are asked to find the dimension of the vector space . So first you have to find a basis of the vector space.
Note:Note: Consider the system of linear equations

$$x + y + z = 0$$
$$z = 0$$

x + y + zz = 0 = 0

Find the solution of the system of linear equations.

Step 1:Step 1:

*1 point*
The system of linear equations has
- ○ unique solution.
- ○ no solution.
- ○ infinitely many solution.

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
Use Gauss elimination method, and observe that $z = 0$z = 0 and $x = -y$.x = −y.
**Accepted Answers:**
infinitely many solution.

*1 point*
Is (1,-1,0) is a solution of the system of linear equations?
- ○ Yes
- ○ No

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
Substitute $x = 1, y = -1$x = 1, y = −1 and $z = 0$z = 0 in both the equations of system of linear equations and check.
**Accepted Answers:**
Yes

*1 point*
Is $(1, -1, 0) \in V$(1, −1, 0) ∈ V ?

○ Yes
○ No

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Yes

Claim:Claim: $\{(1, -1, 0)\}$ spans $V$ .
Step 2:Step 2:
*1 point*
Let $(a, b, c) \in V$ . Which of the following options is/are true?

☐ $a = -b$
☐ $a + b = 1$
☐ $c = 0$
☐ $c = 1$

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
As vector $(a, b, c)$ is a vector of the vector space then it will satisfy the system of linear equations $a + b + c = 0, c = 0$
**Accepted Answers:**
$a = -b$
$c = 0$

Hence, any vector $(a, b, c) \in V$ can be written as $(a, -a, 0)$.
*1 point*
Is $(a, -a, 0)$ a scalar multiple of $(1, -1, 0)$?
○ Yes
○ No

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Yes

*1 point*
If $(a, -a, 0) = \alpha(1, -1, 0)$, then what is the possible value for $\alpha$?
○ $a$
○ $-a$
○ $0$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$a$

Hence, the set $\{(1, -1, 0)\}$ spans the vector space $V$ .

Step 3:Step 3: In this step we have to check whether the set $\{(1, -1, 0)\}$ is linearly independent or not.
*1 point*
Let $S$ be a set which contains only one non-zero vector. Then which of the following options is true?
○
$S$ is linearly independent set.
○
$S$ is linearly dependent set.
**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$S$ is linearly independent set.

Recall:Recall: A basis of a vector space is a set of linearly independent vectors which spans the vector space.

From here you can conclude that $\{(1, -1, 0)\}$ is the basis of the vector space.

Step 4:Step 4: In this step we have to find the dimension the vector space $V$ .

Recall:Recall: The cardinality of the basis is the dimension of $V$ .
Find the dimension of the vector space $V$ .

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*

Check Answers
Your score is: 0/8